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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

LAB WEEK - 4

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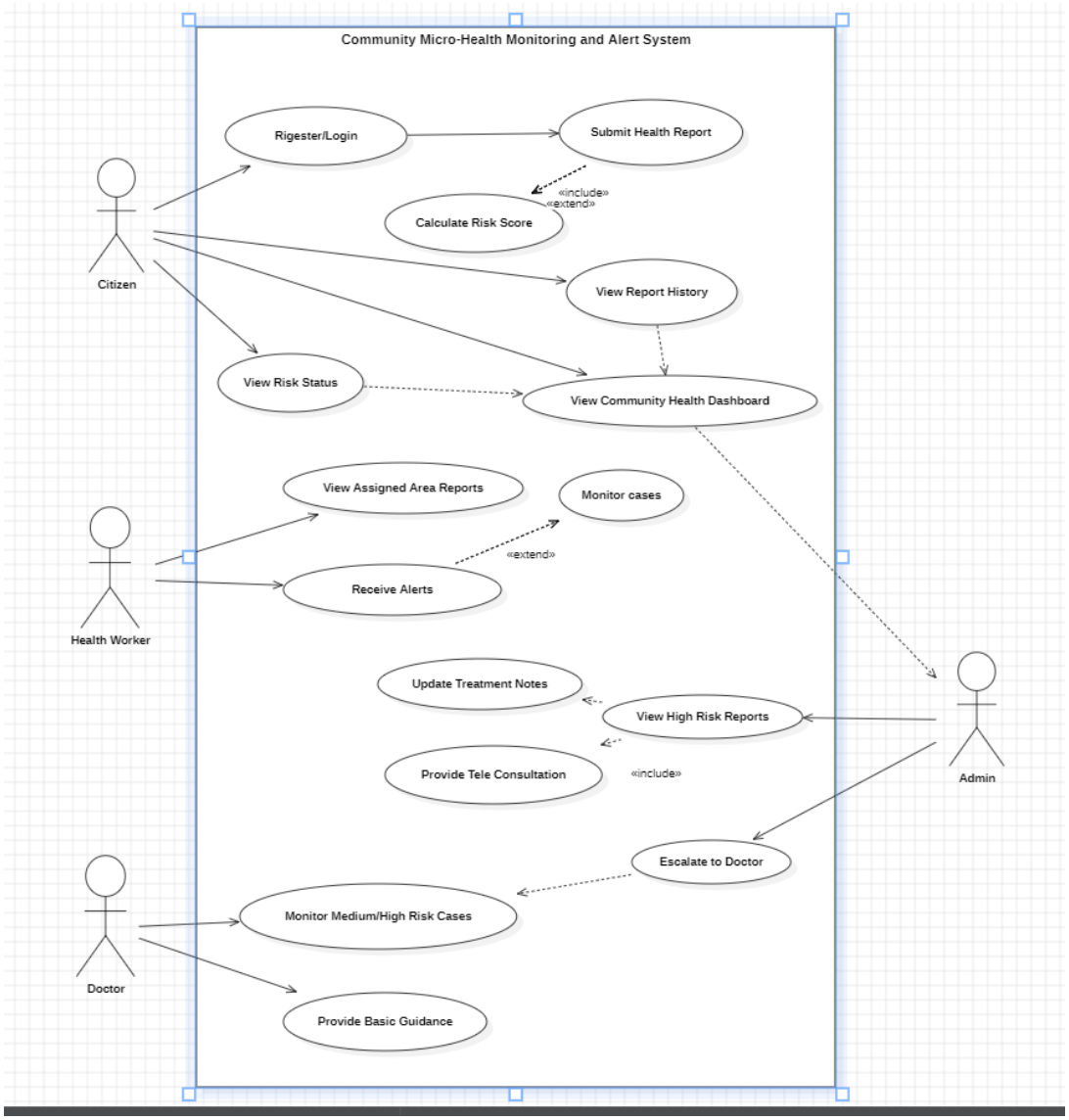
USE CASE DIAGRAM :



- A Use Case Diagram represents the **functional behavior** of the system and shows how external users (actors) interact with the system.
- It helps in identifying **system requirements**, user responsibilities, and system boundaries clearly.
- In the Community Micro-Health Monitoring & Alert System, the main actors are:
    - Citizen
    - Health Worker
    - Doctor
    - Administrator

Actor-wise Functionalities:

- **Citizen**
  - Registers and logs into the system.
  - Submits daily health parameters such as fever, cough, blood pressure, and general discomfort.
  - Views personal health risk status (low, medium, high).
  - Receives alerts and notifications.
  - Requests online consultation with doctors.

- **Health Worker**
  - Logs into the system to access assigned locality data.
  - Receives alerts for medium and high-risk citizens.
  - Responds to alerts and provides guidance.
  - Updates case status and tracks recovery progress.
  - Views community-level health summaries.

- **Doctor**
  - Logs into the system to access patient health reports.
  - Reviews detailed symptom history and risk scores.
  - Provides tele-consultations and medical advice.
  - Tracks high-risk cases for further medical action.

- **Administrator**
  - Manages user accounts and system access.
  - Configures risk rules and thresholds.
  - Maintains system data and records.
  - Monitors overall community health trends using dashboards.
- The Use Case Diagram ensures:
  - Clear **interaction mapping** between users and system.
  - Easy understanding of **system functionality**.
  - Proper definition of **system scope and boundaries**.
  - Better communication between developers and stakeholders.

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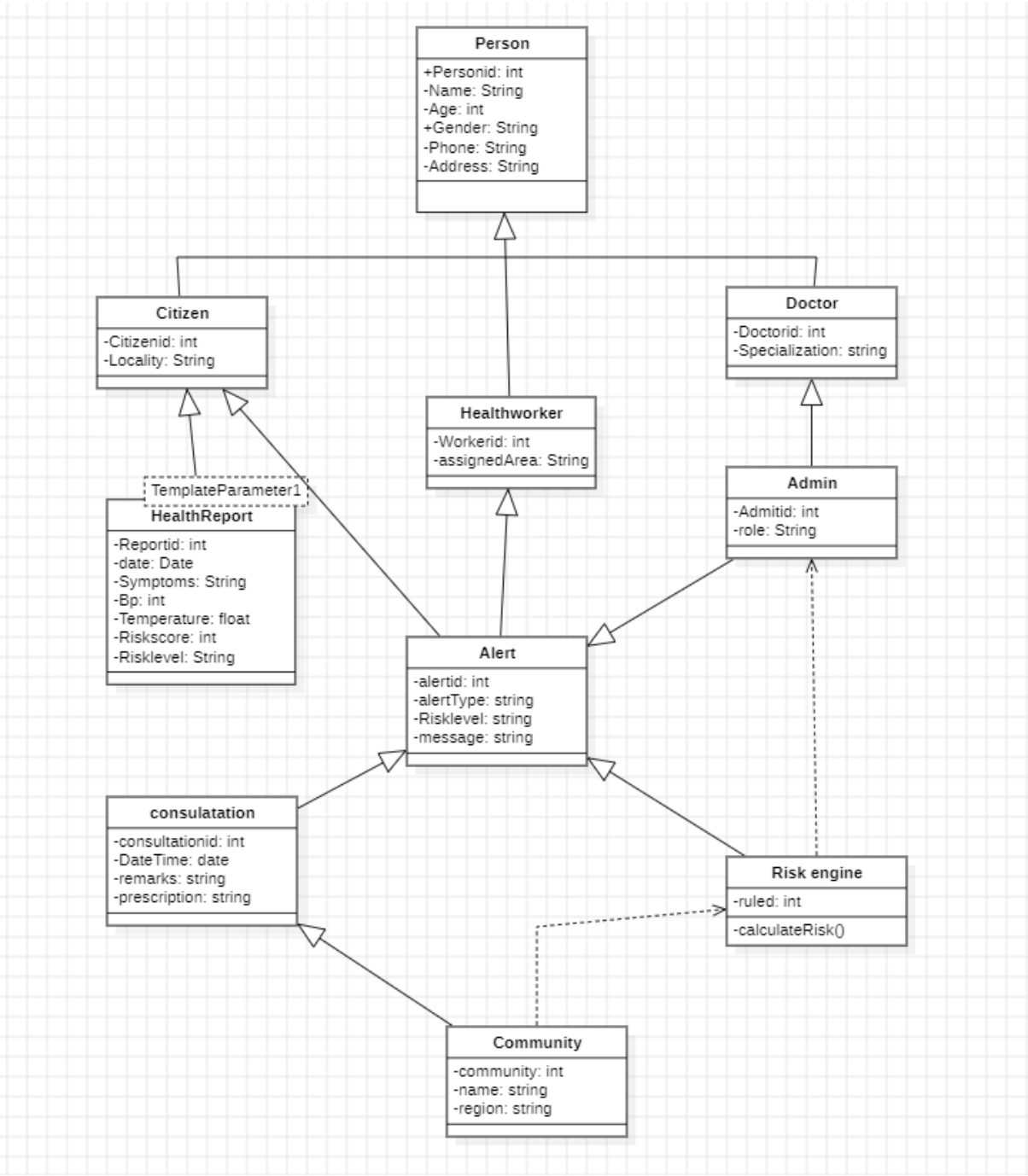
CLASS DIAGRAMS:



- A Class Diagram represents the static structure of the system by showing:
  - Classes
  - Attributes
  - Methods
  - Relationships between classes
- It helps in understanding:
  - How data is organized in the system.
  - How different objects interact internally.
  - How the system can be implemented in code and databases.

Major Classes and Their Roles:

- **Citizen**
  - Stores citizen details such as ID, name, age, contact number, and locality.
  - Allows submission of health data and viewing of risk status.
- **HealthWorker**
  - Stores information about assigned health workers and their service areas.
  - Handles alert responses and case updates.
- **Doctor**
  - Stores doctor information such as specialization and contact details.
  - Manages consultations and monitors critical cases.
- **Admin**
  - Stores administrator credentials and roles.
  - Manages system users, rules, and reports.
- **HealthReport**
  - Stores daily symptom data submitted by citizens.
  - Includes temperature, blood pressure, symptoms, and calculated risk score.
  - Acts as the main data source for alerts.
- **Alert**
  - Stores alert details such as alert ID, risk level, message, and status.
  - Generated when health reports exceed defined thresholds.
- **Consultation**

- Stores consultation records between doctors and citizens.
- Includes consultation date, remarks, and prescriptions.

Relationships Between Classes:

- A Citizen can submit multiple HealthReports (1-to-many relationship).
- Each HealthReport can generate one Alert.
- A HealthWorker can handle multiple Alerts.
- A Doctor can provide multiple Consultations.
- A Citizen can have multiple Consultations.
- Inheritance can be used to share common user details among Citizen, Doctor, HealthWorker, and Admin.

Importance of the Class Diagram:

- Provides a clear blueprint for system implementation.
- Helps in database design and object-oriented programming.
- Improves understanding of system structure and relationships.
- Makes future system modifications easier and more organized.